

Squarefree Masks in the Shift-Two Frame

Exact CRT Fibers, Primitive Truncation Tails, and the Growing-Modulus Ledger

Liang Wang

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Abstract

TPC-127 writes a primitive determinant-two Möbius pair as a shift-two Liouville product on a growing progression, multiplied by the quotient-squarefree mask

$$\mu^2(D(z))\mu^2(V(z)).$$

We resolve this mask without appealing to a density heuristic. For

$$Q_R(m) = \sum_{\substack{k \leq R \\ k^2 | m}} \mu(k)$$

the truncated product has the exact CRT expansion

$$Q_R(D(z))Q_R(V(z)) = \sum_{\substack{k, \ell \leq R \\ (k, s) = (\ell, a) = 1}} \mu(k)\mu(\ell) \mathbf{1}_{z \equiv r_{k, \ell} \pmod{k^2 \ell^2}}.$$

In the TPC-127 variable $n = sV = aD + 2$, the same fiber is the unique solution of

$$n \equiv 2 \pmod{ak^2}, \quad n \equiv 0 \pmod{s\ell^2},$$

and its modulus is $ask^2\ell^2 \leq asR^4$. If the affine values are at most Y on an interval of length N , the total unweighted truncation error is

$$\ll \tau_*(Y) \left(\frac{N}{R} + \sqrt{Y} \right), \quad \tau_*(Y) = \max_{m \leq Y} \tau(m).$$

We also prove the component census $\ll N + R^2$, which replaces a wasteful R^2N triangle bound. These are unconditional finite and elementary estimates. They isolate, but do not prove, the required uniform cancellation on the resulting progression fibers. Thus the paper is L0/L1, not an L2 parity result.

1 Primitive determinant-two setting

Let

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2 \tag{1}$$

be positive on an integer interval I of length N . We use the primitive relations supplied by the literal TPC-93 export,

$$(a, s) = 1, \quad (as, 2) = 1, \quad (d, s) = 1, \quad (u, a) = 1. \tag{2}$$

They imply a, s are odd. Put

$$q = as, \quad n(z) = sV(z) = aD(z) + 2 = su + qz.$$

TPC-127 proves

$$\mu(D(z))\mu(V(z)) = \lambda(q)\lambda(n(z) - 2)\lambda(n(z))\mu^2(D(z))\mu^2(V(z))$$

and transports the literal phase and weight without changing the prefix order [1].

The purpose here is narrower: replace the two squarefree indicators by a finite family of exact congruence fibers and put every truncation cost in a visible ledger.

2 Exact square-divisor expansion

For $R \geq 1$, define

$$Q_R(m) = \sum_{\substack{k \leq R \\ k^2 | m}} \mu(k). \quad (3)$$

The full identity

$$\mu^2(m) = \sum_{k^2 | m} \mu(k) \quad (4)$$

follows prime by prime.

Lemma 2.1 (Compatibility of two square divisors). *For positive integers k, ℓ , the simultaneous conditions*

$$k^2 \mid D(z), \quad \ell^2 \mid V(z)$$

have an integer solution z if and only if

$$(k, s) = 1, \quad (\ell, a) = 1, \quad (k, \ell) = 1. \quad (5)$$

When soluble, they select one residue

$$z \equiv r_{k, \ell} \pmod{k^2 \ell^2}.$$

Proof. If a prime p divides both k and s , then $p \mid D(z)$ and $p \mid s$, hence $p \mid d$, contradicting $(d, s) = 1$. The argument for (ℓ, a) is identical. If $p \mid (k, \ell)$, then p^2 divides both $D(z)$ and $V(z)$, and consequently divides

$$sV(z) - aD(z) = 2,$$

which is impossible.

Conversely, if (5) holds, the first congruence selects one residue modulo k^2 , and the second selects one residue modulo ℓ^2 . Since $(k, \ell) = 1$, the Chinese remainder theorem combines them uniquely modulo $k^2 \ell^2$. $\square$

Theorem 2.2 (Exact truncated CRT expansion). *For every $z \in I$,*

$$Q_R(D(z))Q_R(V(z)) = \sum_{\substack{k, \ell \leq R \\ (k, s) = (\ell, a) = (k, \ell) = 1}} \mu(k)\mu(\ell) \mathbf{1}_{z \equiv r_{k, \ell} \pmod{k^2 \ell^2}}. \quad (6)$$

Equivalently, in the n -frame each nonempty term is the unique residue class solving

$$n \equiv 2 \pmod{ak^2}, \quad n \equiv 0 \pmod{s\ell^2}, \quad (7)$$

with modulus

$$q_{k, \ell} = ask^2 \ell^2 \leq asR^4. \quad (8)$$

Proof. Expand the product in (3) and use lemma 2.1, proving (6). Since $aD = n - 2$ and $sV = n$, the two square divisibilities are exactly (7). The compatibility conditions and $(a, s) = 1$ make the two displayed moduli coprime, so their product is (8). $\square$

Remark 2.3 (No density substitution). The coefficient $\mu(k)\mu(\ell)$, the compatibility conditions, and the actual CRT residue all remain in (6). Replacing them by a squarefree density would not be an identity and would erase the sign needed for reassembly.

3 A primitive truncation tail

Let

$$Y \geq \max_{z \in I} \{D(z), V(z), 2\}, \quad \tau_*(Y) = \max_{1 \leq m \leq Y} \tau(m).$$

Lemma 3.1 (One-form square-divisor tail). *For a primitive positive affine form $L(z) = b + cz$ on I , with $(b, c) = 1$ and $L(z) \leq Y$,*

$$\sum_{z \in I} \left| \mu^2(L(z)) - Q_R(L(z)) \right| \leq \frac{N}{R} + \sqrt{Y} + 1. \quad (9)$$

Proof. By (4),

$$|\mu^2(L(z)) - Q_R(L(z))| \leq \sum_{\substack{k > R \\ k^2 | L(z)}} 1.$$

If $(k, c) > 1$, the congruence is insoluble by primitivity. Otherwise it selects at most $N/k^2 + 1$ integers of I . Since $k \leq \sqrt{Y}$,

$$\sum_{R < k \leq \sqrt{Y}} \left(\frac{N}{k^2} + 1 \right) \leq \frac{N}{R} + \sqrt{Y} + 1.$$

□

Theorem 3.2 (Two-mask truncation bound). *For every complex sequence w_z with $\|w\|_\infty = \max_{z \in I} |w_z|$,*

$$\begin{aligned} & \left| \sum_{z \in I} w_z \left\{ \mu^2(D(z)) \mu^2(V(z)) - Q_R(D(z)) Q_R(V(z)) \right\} \right| \\ & \leq \|w\|_\infty (1 + \tau_*(Y)) \left(\frac{N}{R} + \sqrt{Y} + 1 \right). \end{aligned} \quad (10)$$

In particular the right side is

$$\|w\|_\infty Y^{o(1)} \left(\frac{N}{R} + \sqrt{Y} \right).$$

Proof. Write $A = \mu^2(D)$, $B = \mu^2(V)$, $A_R = Q_R(D)$, and $B_R = Q_R(V)$. Then

$$AB - A_R B_R = (A - A_R)B + A_R(B - B_R).$$

Here $|B| \leq 1$, while

$$|A_R| \leq \sum_{k^2 | D} 1 \leq \tau(D) \leq \tau_*(Y).$$

Apply lemma 3.1 to D and V , multiply by $\|w\|_\infty$, and use the triangle inequality. □

Remark 3.3 (The $\sqrt{Y}$ floor is real for this proof). The $+1$ in the count of each square-divisor progression sums to $O(\sqrt{Y})$. Removing that term uniformly in moving slopes and origins would require additional information. This paper does not hide it in $Y^{o(1)}N/R$.

4 Component census and exponent ledger

Proposition 4.1 (Mass-aware CRT census). *Let $\mathcal{C}_R$ be the sum, over compatible $k, \ell \leq R$, of the number of $z \in I$ in the corresponding CRT fiber. Then*

$$\mathcal{C}_R \leq N \left(\sum_{k \geq 1} \frac{1}{k^2} \right)^2 + R^2 \ll N + R^2. \quad (11)$$

Proof. One residue modulo $k^2 \ell^2$ contributes at most $N/(k^2 \ell^2) + 1$ points. Sum this upper bound and discard the compatibility restrictions. $\square$

The census matters because a uniform estimate relative to each component's *own* absolute mass costs $N + R^2$, not the wasteful $R^2 N$. Physical weights still have to be included in the corresponding weighted census; [proposition 4.1](#) alone does not prove that outer bound.

Suppose, for an exponent audit, that

$$N = X^{\nu+o(1)}, \quad Y \leq X^{\nu+o(1)}, \quad as \leq X^{\kappa+o(1)}, \quad R = X^{\rho+o(1)},$$

and put

$$R_{\text{eff}} = \min\{R, \sqrt{Y}\}, \quad \rho_{\text{eff}} = \min\{\rho, \nu/2\}. \quad (12)$$

Only $k, \ell \leq R_{\text{eff}}$ can occur on the interval. Every retained correlation theorem must be uniform up to modulus exponent

$$\kappa + 4\rho_{\text{eff}}. \quad (13)$$

To state a relative tail exponent, the comparison mass must be declared. Let $\mathcal{V}_X$ be the actual absolute mass of the untruncated literal block and assume

$$\|w\|_{\infty} = X^{o(1)}, \quad \mathcal{V}_X = X^{\omega+o(1)}.$$

If $R < \sqrt{Y}$, [theorem 3.2](#) gives

$$\eta_{\text{tail}} = \min \left\{ \omega - \nu + \rho, \omega - \frac{\nu}{2} \right\}, \quad (14)$$

provided the displayed minimum is positive. In the common mass-comparable case $\omega = \nu$, this reduces to $\min\{\rho, \nu - \nu/2\}$. If $R \geq \sqrt{Y}$, then $Q_R(m) = \mu^2(m)$ for every $m \leq Y$, so the truncation tail is identically zero rather than being governed by [\(14\)](#).

For each retained CRT component c , let $\mathcal{V}_c$ be its actual absolute mass. Define the *weighted census cost* by the theorem-backed inequality

$$\sum_c \mathcal{V}_c \leq X^{\ell_{\text{cen}}+o(1)} \mathcal{V}_X. \quad (15)$$

If the literal weights are uniformly bounded and $\mathcal{V}_X = X^{\nu+o(1)}$, the point census gives the crude choice

$$\ell_{\text{cen}} = (2\rho_{\text{eff}} - \nu)_+. \quad (16)$$

Without those mass hypotheses, [\(16\)](#) is unavailable and ℓ_{cen} must come from the actual weighted archive.

If each CRT component saves $X^{-\eta_{\text{corr}}}$ relative to its own $\mathcal{V}_c$, the certificate returns

$$\eta_{\text{block}}^{\text{cert}} = \begin{cases} \min\{\eta_{\text{tail}}, \eta_{\text{corr}} - \ell_{\text{cen}}\}, & R < \sqrt{Y}, \\ \eta_{\text{corr}} - \ell_{\text{cen}}, & R \geq \sqrt{Y}. \end{cases} \quad (17)$$

The formula is relative to the declared actual absolute mass. It is a loss ledger, not an assertion of the correlation or weighted-census inputs. It is a positive saving only when $\eta_{\text{tail}} > 0$ and $\eta_{\text{corr}} > \ell_{\text{cen}}$ in the first regime, or when $\eta_{\text{corr}} > \ell_{\text{cen}}$ in the exact-cutoff regime.

5 Stop/go rules and claim boundary

TAIL STOP. If $\nu \leq v/2$, the elementary uniform tail (10) gives no positive relative power in the mass-comparable regime while $R < \sqrt{Y}$. One must either prove a stronger weighted tail theorem, choose the exact cutoff $R \geq \sqrt{Y}$, or use a different decomposition. The exact cutoff removes the tail but caps the effective component range at $R_{\text{eff}} = \sqrt{Y}$, with modulus and census costs $\kappa + 2v$ and potentially $(v - \nu)_+$, respectively.

MODULUS STOP. Choosing R large enough to improve the tail inflates the actual modulus to asR^4 . A theorem uniform only for fixed modulus is not callable on these fibers.

CENSUS STOP. The bound (11) is unweighted. If the literal physical weights, the active absolute mass, or outer keys do not satisfy the mass-comparable hypotheses, the actual bound (15) must replace (16).

ARITHMETIC GO. Proceed only with a proved estimate for the actual shift-two Liouville correlation on every retained CRT fiber, with the original phase, weight, origin, and prefix quantifiers. Such a theorem would be genuine L2 input.

Statement	Level and status
Compatibility lemma and CRT expansion	Exact finite theorem (L0).
Primitive tail (10) and census (11)	Unconditional elementary bounds (L0).
Attachment to the TPC-127 actual fixed-2 frame	Literal interface (L1).
Weighted outer census on the complete family	Not proved here.
Uniform cancellation on retained CRT fibers	Not proved; arithmetic gate (L2 if proved).
H3, zero-mode saving, or endpoint theorem	Not proved.

TPC-128 opens the squarefree masks exactly and quantifies the price: tail improvement, modulus growth, and component census compete. Squarefree density by itself supplies none of the required signed correlation. No parity breach, prime-pair lower bound, or twin-prime conclusion is claimed.

References

- [1] Liang Wang. Determinant-Two Liouville Pullback, 2026. Preprint.